WellBot

Your 24/7 Global Wellness Assistant

Milestone: 35% Complete |
Version 0.35.0

Project Overview

About WellBot

WellBot is a full-stack wellness chatbot application designed to provide personalized wellness support 24/7. Built with Flask and modern web technologies, it combines secure user authentication with an interactive chat interface to deliver guidance on mental health, fitness, nutrition, motivation, and stress management.

The application features a beautiful, gradient-based UI with smooth animations and responsive design, making wellness accessible and engaging for all users.

Backend
Flask 3.0 + SQLite

Frontend
HTML5, CSS3, ES6+

Completion
35% Milestone

Security
Password Hashing

Core Features

User Authentication

Secure registration and login with password hashing, session management, and email validation

Interactive Chat

Real-time wellness chat with beautiful UI, quick suggestion chips, and topic-based advice

Database Integration

SQLite database for user management, chat history storage, and wellness data tracking

Modern Design

Gradient color scheme, smooth animations, responsive layout, and intuitive interface

Wellness Topics Covered

Mental Health
Emotional wellbeing support

Fitness
Exercise guidance & motivation

Nutrition
Healthy eating tips & advice

Motivation
Goal setting & inspiration

Stress Management
Relaxation & coping strategies

Frontend: Login & Register

Login Page

- • Username or Email input
- • Secure password field
- • Remember me checkbox
- • Forgot password link
- • Smooth form validation
- • Error message alerts

Register Page

- • Full name (optional)
- • Username (required)
- • Email validation
- • Password strength (min 6 chars)
- • Terms & Conditions checkbox
- • Instant account creation

Frontend Design Details

Color Palette

- Primary: #667eea (Purple Blue)
- Accent: #4facfe (Sky Blue)
- Background: Gradient transitions

Typography

- Font: Poppins (Google Fonts)
- Clean, modern readability
- Consistent sizing hierarchy

Layout

- Split-screen design
- Left: Branding & features
- Right: Forms & authentication

Animations

- Smooth page transitions
- Form input focus states
- Button hover effects

Responsiveness

- Mobile-first approach
- Adaptive layouts
- All devices supported

Branding

- Custom WellBot logo
- Feature icons (□ 💪🥗🎯😊)
- Decorative gradient circles

Technical Architecture

Frontend Layer

HTML5 • CSS3 • JavaScript (ES6+)

Responsive UI with gradient design, form validation, AJAX requests

Backend Layer

Flask 3.0 • Python 3.8+

RESTful API endpoints, session management, password hashing

Database Layer

SQLite 3

User accounts, chat history, wellness data with relational schema

Database Schema

users

- id (PK)
- username
- email
- password_hash
- full_name
- created_at

chat_history

- id (PK)
- user_id (FK)
- message
- response
- category
- created_at

wellness_data

- id (PK)
- user_id (FK)
- mood_score
- stress_level
- sleep_hours
- date

Note: wellness_data table prepared for future tracking features

API Endpoints

POST `/api/register`
Register new user with username, email, password

POST `/api/login`
Authenticate user and create session

POST `/api/chat`
Send message to WellBot and receive wellness advice

GET `/api/chat/history`
Retrieve user's chat history (last 50 messages)

Security Features

- 🔐 Password hashing using Werkzeug security utilities
- 🔑 Session-based authentication with secure cookies
- 🛡 CSRF protection (Flask built-in)
- 💉 SQL injection prevention via parameterized queries
- 🔒 XSS protection through HTML escaping
- ✅ Input validation on both client and server side

Future Roadmap

Remaining 65% Development Plan

Phase 2 (20%)	**Advanced Chatbot** AI/ML model integration • Context-aware conversations • Personalized wellness plans
Phase 3 (20%)	**Wellness Tracking** Daily mood tracker • Sleep & exercise logs • Progress visualization
Phase 4 (15%)	**Social Features** Wellness challenges • Achievement badges • Community engagement
Phase 5 (10%)	**Advanced Features** Export chat history • PDF wellness reports • Mobile app (React Native)

Thank You

WellBot - Your 24/7 Wellness Companion

Milestone: 35% Complete | Active Development